

开源软件仓库与开源软件生态

概念与历史

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Outline

① 开源软件仓库

② 开源生态系统

软件仓库

定义

A software repository is a storage location from which software packages may be retrieved and installed on a computer.

Many software publishers and other organizations maintain servers on the Internet for this purpose, either free of charge or for a subscription fee. Repositories may be solely for particular programs, such as CPAN for the Perl programming language, or for an entire operating system^a.

^ahttps://en.wikipedia.org/wiki/Software_repository

典型软件仓库

软件仓库分类

- 操作系统发行版：基于某种内核的操作系统及外围软件构成的完整系统，如 Debian、Fedora 等 GNU/Linux 发行版，或 FreeBSD、OpenBSD 等 BSD 发行版
- 特定语言仓库：提供针对某个语言的软件包/类库等，如 Perl 的 CPAN、R 的 CRAN、T_EX 的 CTAN、C++ 的 boost 等

软件仓库

软件仓库的操作

Operators of such repositories typically provide a package management system, tools intended to search for, install and otherwise manipulate software packages from the repositories. For example, many Linux distributions use Advanced Packaging Tool (APT), commonly found in Debian based distributions, or yum found in Red Hat based distributions. There are also multiple independent package management systems, such as pacman, used in Arch Linux and equo, found in Sabayon Linux.

APT 梗

今天就能开始用的小知识

```
$apt moo
```

```

          (__)
          (oo)
        /-----\
       /  |       ||
      *  /\----/\
         ~~      ~~

```

```
... "Have you mooed today?" ...
```

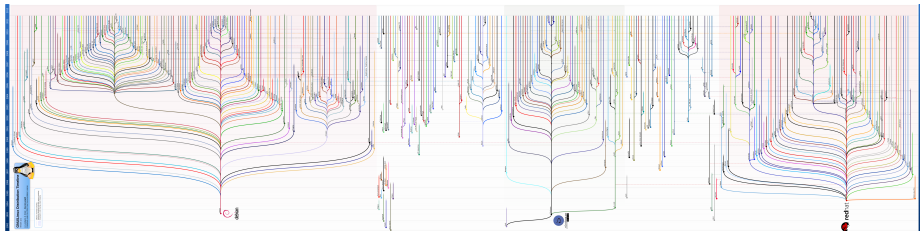
```
$ LC_ALL=C apt |tail -n 1
```

```
$ apt |tail -n 1
```

This APT has Super Cow Powers.

本 APT 具有超级牛力。

Linux 发行版



典型 Linux 发行版

Debian

典型 Linux 发行版

Ubuntu

典型 Linux 发行版

Arch Linux

典型 Linux 发行版

Red Hat

典型 Linux 发行版

Fedora

典型 Linux 发行版

Gentoo

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所谓全栈

Full Stack

To me, a Full Stack Developer is someone with familiarity in each layer, if not mastery in many and a genuine interest in all software technology.

Good developers who are familiar with the entire stack know how to make life easier for those around them. This is why I'm so against silos in the work place. Sure, politics and communication challenges get in the way in large organizations. I think the point Facebook is going for with their hiring policy is, if smart people use their heads and their hearts, a better product gets built in less time^a.

^a<https://www.laurencegellert.com/2012/08/what-is-a-full-stack-developer/>

所谓全栈

Layers of the full stack

- Server, Network, and Hosting Environment.
- Data Modeling
- Business Logic
- API layer/Action layer/MVC
- User Interface
- User Experience
- Understanding what the customer and the business need

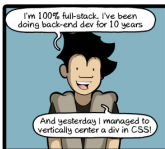
所谓全栈

Which full-stack developer are you?

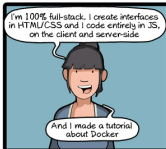
The Full-Stack Devops Engineer



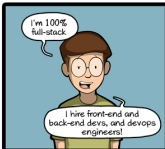
The Full-Stack Back-End developer



The Front-End Full-Stack Developer



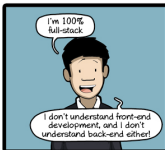
The Full-Stack CTO



The Full-Stack Intern



The Full-Stack Project Manager



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